

Multi-class Text Classification - *Danish Customer Reviews*

Task Description

Objective: Define the goal of automating the classification of Danish customer complaints to reduce manual labor.

Dataset

The dataset consists of 44085 Danish reviews. The dataset consists of review titles, the reviews themselves and a rating between 1-5. The dataset is used for training sentiment classification models. It has been collected from the internet using web scraping techniques. The reviews are written in informal Danish and cover products and services from very different businesses, such as medical, insurance, and goods.

From the original dataset, we filtered for only 1 or 2-star reviews, as these are most likely to contain multiple types of complaints. We then shuffled the dataset to ensure we had different complaint types. We sampled a subset with only 100 reviews which helped to alleviate the work of annotation.

Classes and Labeling Criteria

To ensure consistent and agreeable annotations, we agreed on a definition of each category. Categories utilized for labeling complaints on the sampled reviews:

1. **Credibility** (Troværdighed): Fraud, ethical issues, and aspects that could be discussed in court.
2. **Product quality** (Produktkvalitet): Overall quality of the core business product/service experienced.
3. **Customer service** (Kundeservice): Service provided in person, by phone, or on the Internet to help ensure a successful flow of the business.
4. **Pricing** (Prissætning): Value compared to competitors, transparency of pricing, and price computation.
5. **Delivery issues** (Leveringsproblemer): Correctness of delivery like timing, correct product/service, and other delivery agreements.
6. **Return/refund issues** (Retur og refundering): Functioning of the return and refund processes.

When a review involved more than one category, the above order was used as a priority list when annotating. The annotation was done in Danish.

The resulting annotated dataset has the following class count:

Troværdighed	29
Produktkvalitet	30
Kundeservice	25
Prissætning	7
Leveringsproblemer	7
Retur og refundering	2

Pipeline Design

We created a list of labels with their definitions as a reference for the pretrained models. To provide richer information to the models, we also created a new column by combining the data from the columns `review_title`, `review_text`, and `review_stars`.

```
candidate_labels = [
    "Troværdighed: meaning[Svindel, fup, etiske problemer eller forhold der kan ende i retten.]",
    "Produktkvalitet: meaning[Den generelle oplevelse af kvaliteten af produktet eller kernerydelsen.]",
    "Kundeservice: meaning[Troværdig service ydet personligt, via telefon eller internet for at sikre et godt forløb.]",
    "Prissætning: meaning[Værdi ift. konkurrenter, gennemsigtighed i priser og prisberegning.]",
    "Leveringsproblemer: meaning[Korrekt levering ift. tid, det rette produkt og andre leveringsaftaler.]",
    "Retur og refundering: meaning[Hvordan processen for returnering og tilbagebetaling fungerer.]"]
]

# Join 3 columns info in one
df_labeled['model_input'] = df_labeled.apply(lambda x:
    f"Titel: {x['review_title']}. "
    f"Anmeldelse: {x['review_text']}. "
    f"Kunden gav {x['review_stars']} stjerner.", axis=1)
```

We obtained all models from Hugging Face, and we benefited from the pipelines they provide, which included tokenization and lowercasing.

We worked with the following models

1. Zero-shot:
 - Using NLI (Natural Language Inference): *"joeddav/xlm-roberta-large-xnli"*.
 - Using LLaMA: *"meta-llama/Meta-Llama-3.1-8B-Instruct"*
2. Fine-tune on 2 pre-trained transformer model
 - Using: *"google-bert/bert-base-multilingual-cased"*
 - *"Maltehb/danish-bert-botxo"*

We had to use 4-bit quantization with LLaMA to fit it within our available 8GB of GPU memory. In hopes of yielding better results and gaining more experience working with large language models, we tested the Zero and Few Shot classification with Qwen and Phi-3 as well; however, LLaMA produced better results in both cases.

Training and Evaluation

Early in the training phase, we were met with lower-than-acceptable accuracies, which led us to seek improvements through tweaking of the samples. We experienced improvements when the labels were together with their explanation. We observed a difference when using the new column (concatenated from the other 3 columns) versus feeding the model only the review_text. The latter was better with fine-tuning BERT. Due to the class imbalance, we noticed that BERT was taking the ‘lazy’ approach of achieving better results by predicting only the most common classes. Therefore, we decided to try oversampling, and, with the dataset balanced as follows, we achieved an important improvement.

Prissætning	22
Produktkvalitet	22
Kundeservice	22
Leveringsproblemer	22
Troværdighed	22
Retur og refundering	22

Results and Analysis

Model Comparative Table:

	Precision	Recall	F1 Score	Accuracy
NLI Model	0.6392	0.4125	0.3753	0.49
LLM Zero-Shot	0.3825	0.3731	0.2821	0.42
LLM Few-Shot	0.7379	0.6702	0.6487	0.66
Bert Finetune English	0.3152	0.4540	0.3711	0.48
Bert Finetune Danish	0.4074	0.4471	0.3389	0.48

We observe that LLM (LLaMA) Few-Shot was clearly best. We can understand that a less complex NLI model would not be the best (though its F1 Score was not the worst).

Nevertheless, we were expecting better results from fine-tuning BERT. We believe that 100 samples are too few for effective fine-tuning. By investing more time in annotating a significantly larger number of samples, BERT will likely be better or as good as LLaMA.

An unexpected result was that the Danish BERT did not perform better than the English one. We’ve concluded that this may be due to the lack of high-quality and diverse training on the language, in comparison to English. The nature of the dataset is also very informal, which makes classification harder.